

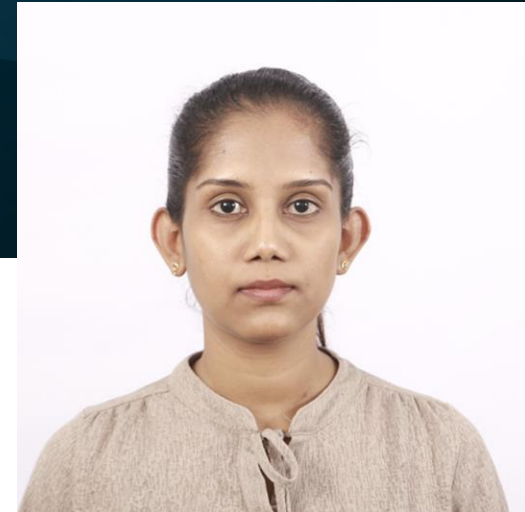
An Adaptive Intelligence Framework for Data-Driven Optimization of Collaborative Software Project Environments

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An Adaptive Intelligence Framework for Data-Driven Optimization of Collaborative Software Project Environments



Supervisor: Prof. Samantha Rajapaksha

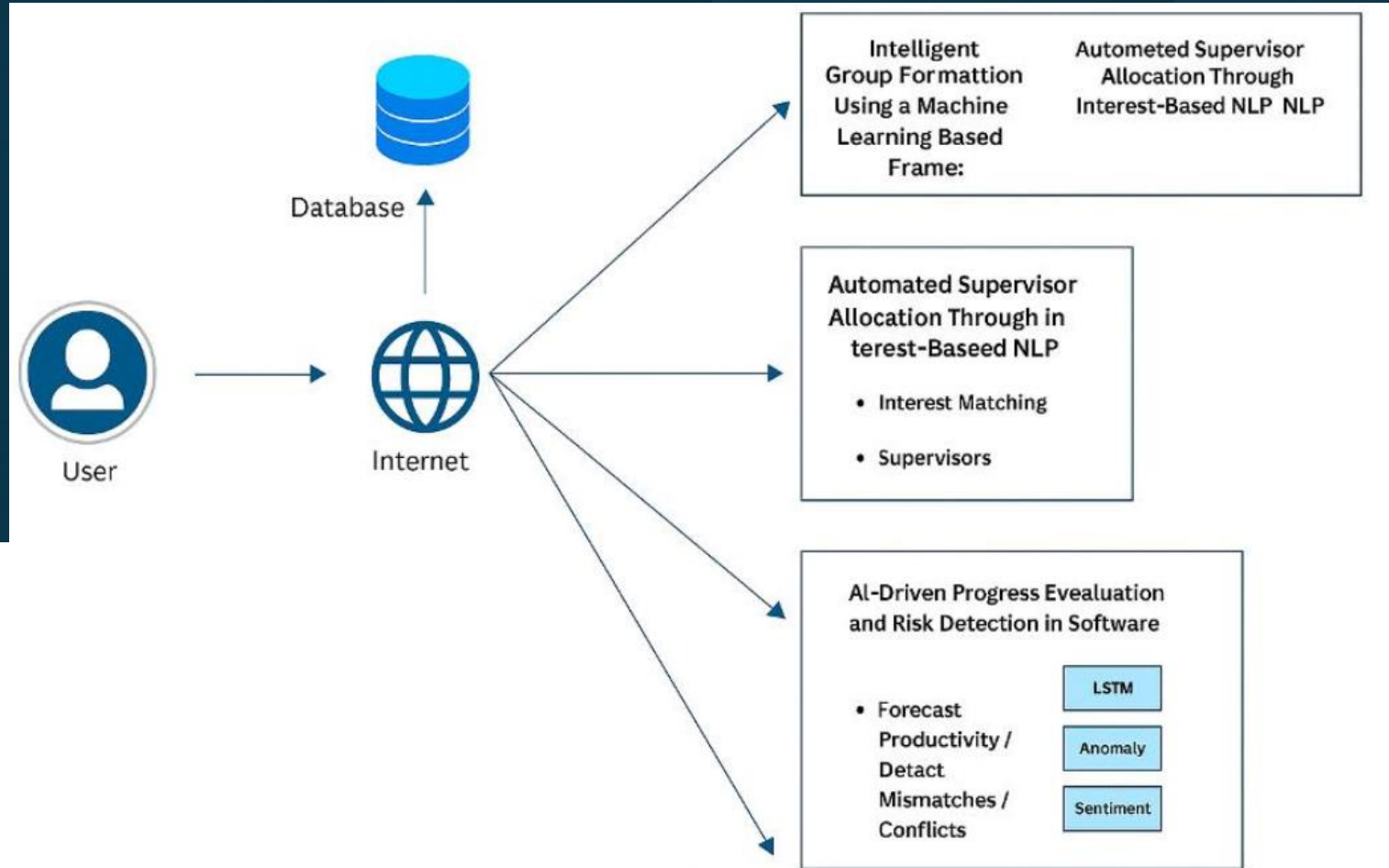


Co-Supervisor: Ms. Thamali Kelegama

An Adaptive Intelligence Framework for Data-Driven Optimization of Collaborative Software Project Environments



An Adaptive Intelligence Framework for Data-Driven Optimization of Collaborative Software Project Environments



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INTELLIGENT GROUP FORMATION AND TOPIC EVALUATION USING LLM

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INTELLIGENT GROUP FORMATION AND TOPIC EVALUATION USING LLM

Solutions

Automatically forms balanced student groups using ML

1

Uses TF-IDF to analyze student interests and skills

2

Applies K-Means clustering for intelligent grouping

3

Ensures equal group size (4 students) with balancing logic

4

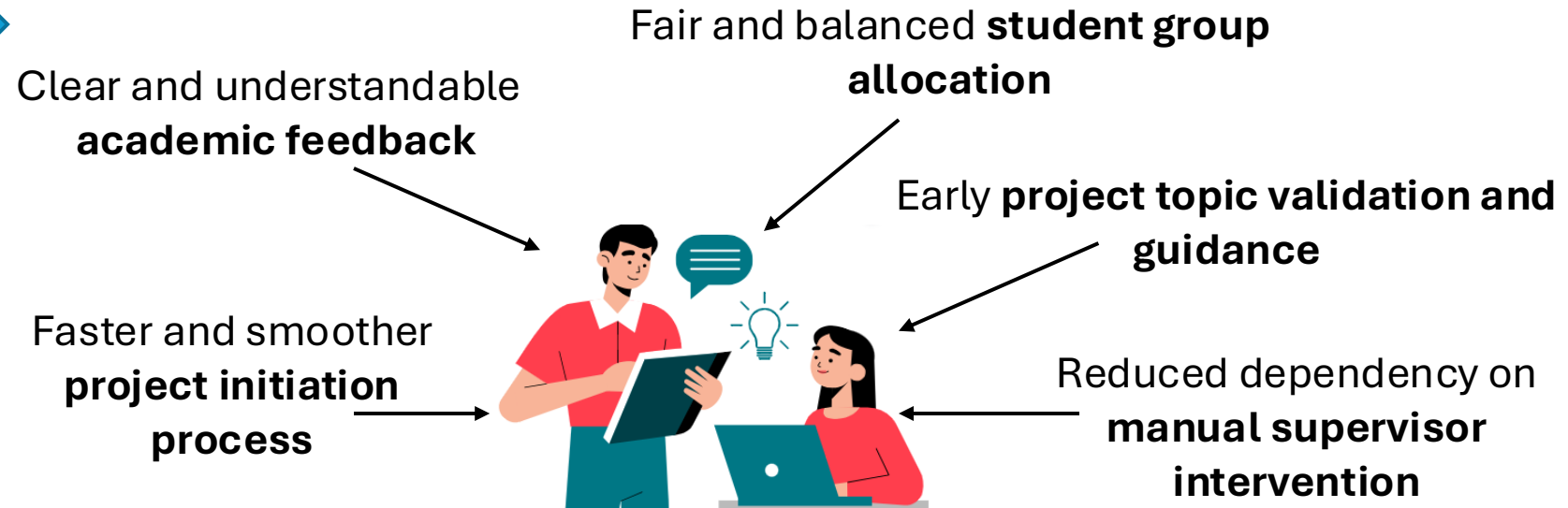
Evaluates project topics early using LLM-based analysis

5

Generates structured and actionable feedback

6

User Requirements Addressed by the Solution



Sub-problem:

Manual group formation results in imbalanced teams and free-rider issues, reducing collaboration quality. Students often propose unclear or infeasible project topics due to the lack of early academic guidance. This increases supervisor workload through late-stage corrections and leads to repeated rework, project delays, and higher student stress.

INTELLIGENT GROUP FORMATION AND TOPIC EVALUATION USING LLM

Design Excellence / Contribution

- End-to-end **AI-driven group formation pipeline**
- Combines **numerical + textual features** for better grouping accuracy
- Integrated **topic evaluation within the same workflow**
- Provides **explainable and transparent outputs**
- Scalable and reusable for **large student cohorts**



TO DO LIST

- Create **visual dashboards** for group composition
- Add **Frontend–Backend** Integration for End-to-End System
- Improve **structured LLM feedback format**
- Enhance **explainability of grouping decisions**

User Feedback on Prototype

- Groups were **more balanced** compared to manual allocation
- Topic feedback was **clear and easy to understand**
- Early guidance **reduced rework and proposal delays**
- Improved student **confidence in team and topic selection**

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SUPERVISOR-STUDENT MATCHING AND NLP- BASED PROJECT GUIDANCE SYSTEM

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SUPERVISOR-STUDENT MATCHING AND NLP- BASED PROJECT GUIDANCE SYSTEM

Solutions

Provides NLP-based supervisor-student matching

1

Matches student project topics with supervisor expertise

2

Ensures fair supervisor workload distribution

3

Provides continuous AI-based project guidance through a chatbot

4

Improves mentoring quality and topic relevance

5

User Requirements Addressed by the Solution

Improved understanding of project direction

Accurate supervisor-topic alignment

Continuous support beyond physical meetings

Personalized academic guidance

Reduced dependency on manual allocation



Sub-problem:

Students are often assigned supervisors manually, based mainly on availability rather than expertise. This leads to topic-supervisor mismatch, limited guidance outside meetings, and reduced project quality. Existing systems lack semantic understanding of project topics and real-time AI mentoring support.

SUPERVISOR-STUDENT MATCHING AND NLP- BASED PROJECT GUIDANCE SYSTEM

Design Excellence / Contribution

- Designed a **NLP-based semantic matching pipeline** instead of **manual** supervisor allocation.
- Implemented **similarity-based** ranking to recommend the most suitable supervisors per project.
- Integrated an AI chatbot to provide continuous, context-aware project guidance.
- Ensured scalable and fair allocation by considering supervisor **availability** and **expertise**.

TO DO LIST

- Improve supervisor profile enrichment
- Add feedback-based recommendation refinement
- Enhance chatbot responses with domain-specific data
- Visualize supervisor matching confidence scores

User Feedback on Prototype

- Supervisor recommendations were more relevant
- Improve clarity on project direction
- Reduced confusion during early project stages
- Continuous guidance improved confidence and engagement

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AI-DRIVEN PROGRESS EVALUATION AND RISK DETECTION

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AI-DRIVEN PROGRESS EVALUATION AND RISK DETECTION

Solutions

Provides AI-based continuous project progress monitoring

1

Uses LSTM models to forecast future project performance and delays

2

Applies sentiment analysis to evaluate team engagement and collaboration quality

3

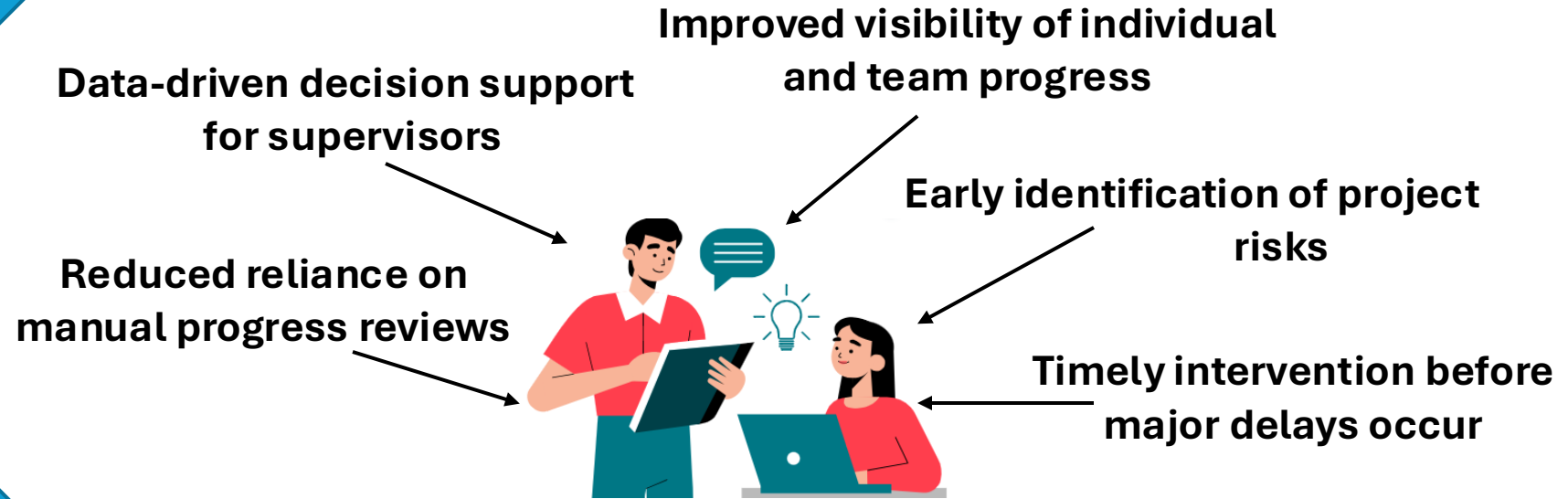
Detects early risks using anomaly detection techniques

4

Delivers real-time dashboards and risk alerts for supervisors and students

5

User Requirements Addressed by the Solution



Sub-problem:

Students' project progress is often monitored manually through periodic meetings and reports, making it difficult to detect risks early. Supervisors lack real-time insights into student engagement, productivity drops, or missed milestones. As a result, issues such as delays, low participation, and team conflicts are identified too late, leading to poor project outcomes and increased stress for both students and supervisors.

AI-DRIVEN PROGRESS EVALUATION AND RISK DETECTION

Design Excellence / Contribution

- Designed an **AI-driven progress evaluation framework** instead of manual milestone tracking.
- Implemented **LSTM-based time-series forecasting** to predict project delays and performance trends.
- Integrated **sentiment analysis** to assess team morale, engagement, and collaboration quality.
- Applied **anomaly detection** to automatically flag unusual behaviors such as inactivity or productivity drops.
- Developed **real-time dashboards** to visualize risks, progress trends, and predictive insights.

TO DO LIST

- Improve LSTM prediction accuracy with more data
- Enhance sentiment analysis using academic domain data
- Add explainable risk scores for supervisors
- Introduce automated intervention recommendations

User Feedback on Prototype

- Improved visibility of project progress and team performance
- Early risk alerts helped supervisors intervene on time
- Reduced uncertainty and stress during project execution
- Dashboards made evaluation more transparent and objective

AI-DRIVEN PROGRESS EVALUATION AND RISK DETECTION

Customer Persona

- Final-year undergraduate students
- Academic supervisors
- Universities and faculties

Market Size

- Large number of final-year group projects every year
- Manual group formation causes imbalance and free-rider issues
- Students submit unclear or infeasible project topics
- Supervisors face high workload and late-stage corrections
- Lack of early guidance leads to rework, delays, and stress

What Is Unique in This Solution

- AI-based intelligent group formation using machine learning
- NLP-based analysis of student interests and project topics
- Early topic evaluation using LLMs before supervisor review
- Structured academic feedback (clarity, feasibility, risks)
- Interest-based supervisor–student matching
- AI-driven progress monitoring and early risk detection

How Will the Cost Be Recovered

- Institutional subscriptions (universities & faculties)
- Premium academic analytics and mentoring features
- Free basic version for students



THANK YOU!!